



Knowing Myself semester 1

Name:

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Essential Skills and Personal Qualities

These are skills I have evaluated what I have developed during the first semester of university and which do I need to strive to develop/ improve.

- **Problem Solving and creativity** – the ability to find solutions to problems through using initiative, skills of analysis and reasoning.
- **Teamwork** – the ability to cooperate with others to meet shared goals, to consider others' opinions and support team colleagues to achieve objectives.
- **Written communication** - the ability to write concisely and convey meaning in a manner appropriate to different readers.
- **Verbal communication**– the ability to verbally express your ideas clearly and persuasively and to adapt what you say according to the situation and audience.
- **Willingness to learn** – the ability to take responsibility for your learning, seek new learning opportunities, and respond well to feedback.
- **Organisation and planning** – the ability to plan and organise your workload and to prioritise tasks accordingly.
- **Leadership** – the ability to take control of a situation and to inspire and motivate others to meet goals.
- **Drive and motivation** – the ability to see something through despite any setbacks and to display enthusiasm and a positive outlook.
- **Business awareness** – the ability to demonstrate that you have an understanding of how a business operates, what products or services it offers, who its competitors and customers are and what issues it is facing in the market.

Section 1. REFLECTING ON YOUR STUDIES

Learning from your four semester A modules

Module Name: ECS401U procedural programming

1. What technical skills did you develop or acquire from this module?

This module focused on developing my critical thinking abilities in terms of programming targeting thinking to solve a problem in a procedural way using decomposition. It also developed my understanding of the basics of programming in java not just how to write code but the basic ideas of what is going on under the hood building a platform for later learning.

2. Describe each piece of coursework (e.g. essay, report, lab exercises, group based or individual project) that you completed for this module and consider the following questions:

- a) What skills did you develop or acquire as a consequence of doing this coursework?

The coursework for the module was 7 shorts (small programs targeting competence) and a mini project (integrating all those competencies into one project). The shorts where 'short' programs that targeted a programming construct or idea learned previously in the module and help reinforce the concept and build the problem-solving skills and meetings deadlines for the work. Also, there was an end-term test as part of the coursework in which we had to write the answers on paper ensuring deep understanding of the content and language basics to ensure you can get a high mark.

- b) What evidence do you have of the work that you did that you could show to other people? This could include examples of code, written reports/essays, copies of presentation slides, URL links to a project or to a GitHub account...

<https://github.com/EScott-LL10T/Eurovision-song-contest-project-procedural-programming> is the link to my Eurovision song contest program that was my mini project however I didn't make copies on the shorts as we developed them on Jupiter hub, and they got scrubbed from the server. However, this is a lesson I will use to remember to save my work and make copies/ backups in the future.

3. What do you know now that you didn't know before you took this module?

I now know the procedural way to create a program; the core basics of java that can help me understand the language better going forward as I work with it in future semesters and those crucial core basics that any applies to any programming language.

Module Name: ECS404U Computer system and networks

1. What technical skills did you develop or acquire from this module?

This module targeted the basics of computer systems targeting the architecture of the computer covering topics like transistors, binary to character representation. Then for the network side of things, we covered Wireshark and learned how to search through and understand the data packets and, we learned the basics of the OSI model and protocols before that. I wouldn't say there was any one skill I learned from the module except willingness to learn perhaps due to ensuring I understood a topic as the content of the module was difficult in my opinion except from the assembly language taught under the computer architecture umbrella and help develop problem solving skills.

2. Describe each piece of coursework (e.g. essay, report, lab exercises, group based or individual project) that you completed for this module and consider the following questions:

a) What skills did you develop or acquire as a consequence of doing this coursework?

There was two types of coursework. One was in form of lab quizzes which helped apply the content we had learnt in the lectures. They didn't necessarily develop any skills however they did teach me the importance of application to aid learning and ensure understanding. Then the second coursework was assembly language, and this helped develop problem solving skills.

- b) What evidence do you have of the work that you did that you could show to other people? This could include examples of code, written reports/essays, copies of presentation slides, URL links to a project or to a GitHub account...

Example of code:

```
# store output pointer in $t0
move $t0, $a1

##### DNS header construction
#####
# create the 12 bit DNS header in big endian.

# transaction ID

# $t1 = $a2 * 2^8
srl $t1, $a2, 8

# 0($t0) = $t1
sb $t1, 0($t0)

# 1($t0) = $a2
sb $a2, 1($t0)

# flags

# $t2 = 0x01
li $t2, 0x01

# 2($t0) = $t2
sb $t2, 2($t0)
```

```

# 3($t0) = 0
sb $0, 3($t0)

# Questions (QDCOUNT)

# 4($t0) = 0
sb $0, 4($t0)

# $t2 = 1
li $t2, 1

# 5($t0) = $t2
sb $t2, 5($t0)

# Answer RRs (ANCOUNT)

# 6($t0) = 0
sb $0, 6($t0)

# 7($t0) = 0
sb $0, 7($t0)

# Authority RRs (NSCOUNT)

# 8($t0) = 0
sb $0, 8($t0)

# 9($t0) = 0
sb $0, 9($t0)

# additional RRs (ARCOUNT)

# 10($t0) = 0
sb $0, 10($t0)

# 11($t0) = 0
sb $0, 11($t0)

# $t0 = $t0 + 12
addiu $t0, $t0, 12

# $t6 becomes the input string pointer
move $t6, $a0

##### QNAME Encoding Algorithm
#####
# outer loop

# branch to encode
b .encode

.encode:
# move $t0 to $t3 to reserve space for length
move $t3, $t0

```

```

# $t0 = $t0 + 1
addiu $t0, $t0, 1

# $t4 = 0 ($t4 is the length counter of the packet in bytes)
li $t4, 0

# branch to copyAndCount
b .copyAndCount

# inner loop

.copyAndCount:

# $t5 = 0($t6)
lb $t5, 0($t6)

# if $t5 == 0, branch to endLabel
beq $t5, $0, .endLabel

# if $t5 == 46 branch to endLabel
beq $t5, 46, .endLabel

# $t5 = 0($t0)
sb $t5, 0($t0)

# $t0++
addiu $t0, $t0, 1

# $t6++
addiu $t6, $t6, 1

# $t4++
addiu $t4, $t4, 1

# branch to copyAndCount
b .copyAndCount

.endLabel:

# $t4 = 0($t3)
sb $t4, 0($t3)

# $t5 = 0($t6)
lb $t5, 0($t6)

# branch if $t5 == 0 to finish_qname
beq $t5, $0, .finish_qname

# $t6++
addiu $t6, $t6, 1

# branch to encode
b .encode

```

```

.finish_qname:

# 0($t0) = 0
sb $0, 0($t0)

# $t0++
addiu $t0, $t0, 1

# 0($t0) = 0
sb $0, 0($t0)

# $t2 = 1
li $t2, 1

# 1($t0) = $t2
sb $t2, 1($t0)

# $t0 = $t0 + 2
addiu $t0, $t0, 2

# 0($t0) = 0
sb $0, 0($t0)

# 1($t0) = $t2
sb $t2, 1($t0)

# $t0 = $t0 + 2
addi $t0, $t0, 2

# $v0 = $t0 - $a1
subu $v0, $t0, $a1

```

This was my code for the part d of my coursework I also have the files of all the other parts however this was the hardest one. The other coursework I have no evidence of as they were just quizzes.

3. What do you know now that you didn't know before you took this module?

I now know and understand the basics of computer architecture and a greater understanding of networks and how data packets are sent over a network, also the layout of a standard network model (OSI model). I also understand the basics of MIPS assembly language.

1. What technical skills did you develop or acquire from this module?

This module covered box proofs and sets, so therefore more theory based/ mathematical module compared to the other three. However, it does prepare me for certain data structures and a way of thinking that can help programming and more in a general computer science way.

2. Describe each piece of coursework (e.g. essay, report, lab exercises, group based or individual project) that you completed for this module and consider the following questions:

- a) What skills did you develop or acquire as a consequence of doing this coursework?

There were three coursework quizzes that were completed before a deadline and apart from that there was only a mid-term test. Mainly the coursework helped reinforce and practice applying what you have learned by answering questions.

- b) What evidence do you have of the work that you did that you could show to other people? This could include; examples of code, written reports/essays, copies of presentation slides, URL links to a project or to a GitHub account...

For this module there is no way to document the coursework.

3. What do you know now that you didn't know before you took this module?

I learned how to do proof by natural deduction using box proofs and the basics of sets and graphs.

1. What technical skills did you develop or acquire from this module?

In this module, it covered more about the ethics side of computer science and more specifically gen ai and then covered scientific method for evaluating a project and what goes into picking a problem to resolve and the process that goes into it. This module greater improved my communication and teamwork skills

2. Describe each piece of coursework (e.g. essay, report, lab exercises, group based or individual project) that you completed for this module and consider the following questions:

- a) What skills did you develop or acquire as a consequence of doing this coursework?

The coursework was in the form of two group projects. One was a presentation on gen ai and 2 different facets of it and the pros and cons and concluding how it affects those facets. The second group project was defining a project of choice of 11 or so and as a group we would research different parts of a project depending on competencies of the project until the whole project and layout was complete.

- b) What evidence do you have of the work that you did that you could show to other people? This could include; examples of code, written reports/essays, copies of presentation slides, URL links to a project or to a GitHub account...

<file:///C:/Users/ellio/Downloads/PRP-A1-05.05-slides-1.pdf>

is the link to the slides for my group project for the first assignment.

<file:///C:/Users/ellio/Downloads/Group05.11 ECS427U assignment2-.pdf>

is the link to the pdf to the second assignment.

3. What do you know now that you didn't know before you took this module?

The strengths and weaknesses of gen ai and the process of using a scientific method and the research that goes into defining a project before a single part of it has even been made.

Looking back on your first year so far

1. What subjects/topics are capturing your interest and arousing your curiosity *and why?*

The main topics/ subjects that captured my interest was procedural programming as I really enjoy the problem solving and the creativity that comes with programming. I also really enjoyed box proofs and natural deduction as I really didn't understand it at first, but once I spent the time understanding it, I really enjoyed the process of answer questions via box proofs.

2. How have your skills/knowledge/personal qualities improved since you started your course?

I have become more hard working since starting the course and punctual as you must meet deadlines and make sure you show up on time to lectures to not get left behind. I also feel I have greatly improved my problem solving skills.

Section 2: REFLECTING ON YOUR EXPERIENCES

All experience counts and the activities you take part in in your life tell a story about you and provide evidence of a range of different skills.

Your experiences may include: paid work, working for a family business, volunteering, supporting your community or faith group, hobbies, sports, student societies and any side projects or learning you do in your spare time.

Identify a minimum of 3 experiences/activities that you take part in or have done in the past and answer the questions below:

Experience/ Activity 1

1. Describe the activity (what was it and what were you doing?)

Normally, I watch a lot of youtube/ social media but all of it is pretty useless/ for comedy. However, youtube can be a very resource if you find the correct channel. Then I found Bro Codes and started watching his tutorials to learn other programming languages or to help if I am stuck with java for example.

2. What specific skills have you gained from this experience?

In terms of technical skills, I would say I have improved my general understanding of programming at a basic level. Also, I have learned to use resources outside of my degree and am going forward to helping me teach things rather than just struggling.

3. What else have you learned about yourself from this experience?

I have learned that if I don't understand something it's not because I'm dumb etc but I just need to expose myself to it long enough then I can learn anything.

Experience/ Activity 2

1. Describe the activity (what was it and what were you doing?)

One of my hobbies is to go and watch football (specifically Charlton athletic) and when I go to watch a game I usually make an effort to chat to the people sat next to me as since we have common ground (we like the same team) we can get along and create a positive experience even if the football that day doesn't pan out.

2. What specific skills have you gained from this experience?

This helps with my communication skills and how I will have to get along with potential colleges as even though they may not necessarily enjoy the job as I do football, but they are in the same situation as me in the sense that allowing me to be able to get along with them.

3. What else have you learned about yourself from this experience?

I have learned that I can be quite extroverted once an established ground or commonality is established and I can get along with people well if I am nice.

Experience/ Activity 3

1. Describe the activity (what was it and what were you doing?)

Going to the gym. I have been going to the gym on and off for too long however it is something I enjoy when I go.

2. What specific skills have you gained from this experience?

I believe I have gained persistence and patience going to the gym as when you are working out you must work hard but for the whole session. However, you can just do one exercise then go straight to another you have to wait as you can do it all at once hence patience being crucial.

3. What else have you learned about yourself from this experience?

I have learned persistence, patience and consistency are the keys to success in the gym if not in life, however consistency is the hardest of the three as you have to follow the other 2 guides over a long period of time.

Section 3: REFLECTING ON YOUR WELLBEING

Hopefully, most of your time spent as a student in the School of EECS will be enjoyable, stimulating and fulfilling. However, sometimes you may encounter stressful and challenging situations that can have an impact on your wellbeing and your studies. Being able to recognise why this has happened and what options you have available to mitigate the impact will make it easier for you to manage future challenges as they arise.

1. What have you found most challenging about being a university student so far?

The hardest part of university student so far for me is understanding difficult concepts and if you don't you feel you are so behind of work if you take the time to understand the concept, even if you aren't increasing cortisol.

2. What impact did this have on your sense of wellbeing?

In the immediate I felt stressed, however over time I realised I wasn't behind and then I respond very positively. Also, the stress almost increases my productivity, so it was a net positive still.

3. What have you tried so far to help you deal with this? *(this could include: getting advice from other people, implementing new study skill habits or time management practices, factoring in time for socialising/leisure activities, self-care and so on.)*

I try to keep social, do a hobby, or take the problem I am dealing with head normally sorts it out straight away.

4. What do you think you could do differently in the future to help you to deal with any challenges you might face?

Positive reinforcement and just try to complete work early to avoid the feeling altogether.

Don't forget that the University is here to support you so please don't hesitate to let us know if you need any help or advice at any time.

QM Advice and Counselling: <https://www.qmul.ac.uk/welfare/>

Academic Study Support: <https://www.qmul.ac.uk/library/academic-skills/>

Section 4: ABOUT ME

Your LinkedIn profile should include a summary (also known as 'Profile' or 'Career Objectives') of who you are and what makes you tick.

This section should be:

- A short paragraph of no more than 6 sentences in length.
- You can include the following -
 - o What you study;
 - o Your interests and what you do in your spare time;
 - o What motivates and excites you;
 - o Any potential aspirations / ambitions (if you have any career, personal goals etc.).

To help you to create an 'About Me' section answer the following questions:

1. What areas of technology are you passionate about?

I am particularly passionate about programming even though I haven't got the knowledge yet, but I think software development is the most interesting to me now.

2. What do you consider are your main strengths and skills (these can be technical and non-technical)?

My main strengths are my persistence and my willingness to learn.

3. Outside of technology what other things interest or excite you?

Outside of technology, what excites me is mainly my hobbies, probably music and football, however my main drive is to make my parents proud.

Now have a go at writing your 'About Me' paragraph:

I study computer science at QMUL, and I am most interested in software development. This is because I love solving problems and feel like climbing a mountain: the 10 hours of climbing is worth it just to see the top for those 30 minutes you are there. I personally think my main strengths are my persistence and my willingness to learn and engage with the stuff I truly enjoy. My main drive is to be able to enable myself to enjoy my hobbies even in adulthood and to make my family, present and future proud.